Field Remote Sensing Explorer

What the tool does with a field boundary -- steps, data and outputs

Overview, November 2026. Example field: Story County, Iowa (65.9 ha, corn-soybean rotation, 2020-2026). Every figure here is the tool's own output for that field.

You give the tool one field boundary. It reads free satellite and public data **only inside that boundary**, turns it into a per-season management history -- what was grown, when it was planted and harvested, whether the soil was covered over winter -- and shows it in nine app tabs plus a summary table. No field visit and no grower input are used. The same pipeline runs in an R (Shiny) and a Python (Streamlit) app; features marked *Python only* are not yet in the R app.

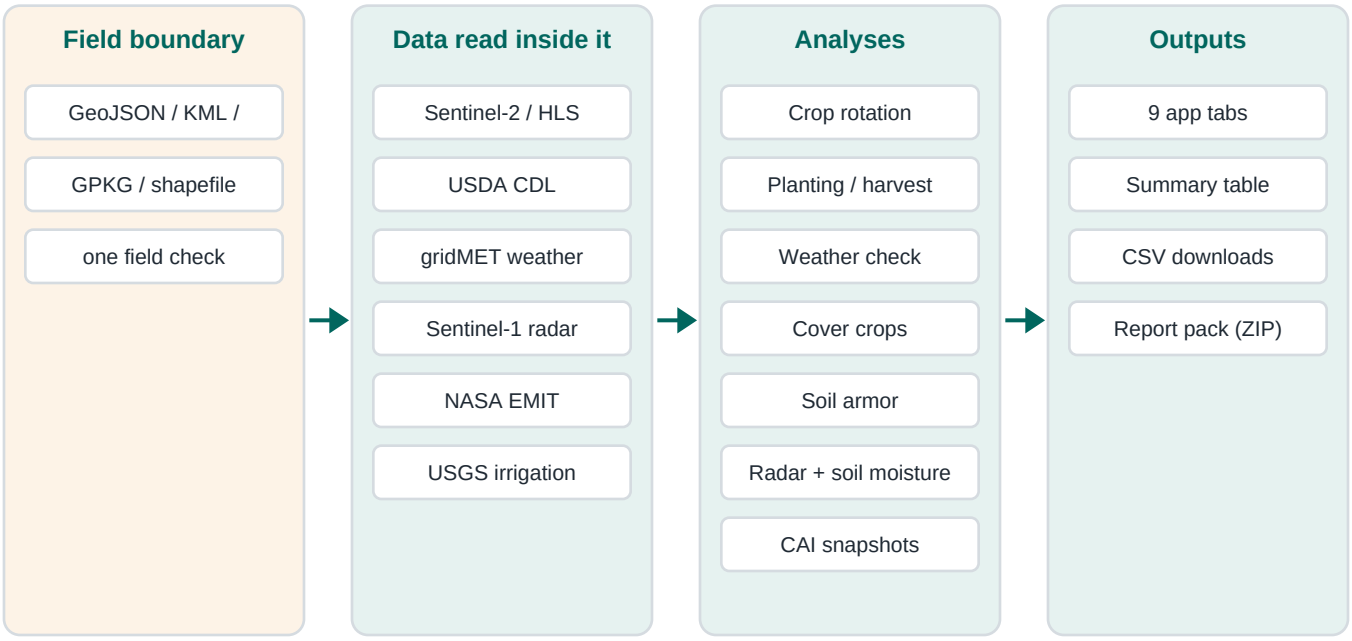


Figure 1. The pipeline. Each data source is read for the boundary only; the analyses combine them; everything ends in the app tabs and the summary table.

At a glance

What	How	Status
Crop each year	USDA Cropland Data Layer, dominant class inside the boundary	Published product
Planting and harvest dates	Sentinel-2 NDVI curve per season, crop-specific lags, uncertainty window from observation gaps	Not yet compared to grower dates
Field-work dates	NDVI dates moved off rain / wind days (gridMET); radar and soil moisture narrow the window	Prototype
Cover crops	Off-season NDVI between harvest and next planting	Thresholds wrong outside the Midwest
Soil armor	Living / residue / bare fractions from NDVI and a shortwave index, before planting and per calendar year	Uncalibrated
Crop residue (CAI)	NASA EMIT hyperspectral snapshots (Python only)	Prototype, uncalibrated

The steps, and what each does to the boundary

#	Step	What happens inside the boundary	Data	App tab
1	Load and check the boundary	Accepts GeoJSON, KML, GeoPackage or zipped shapefile; keeps one field (slivers dropped, two fields refused). Checks with CDL whether the outline mixes two fields and how much of it is not cropland.	Boundary, CDL	Field
2	Data availability	Lists every Sentinel-2 / Landsat / Sentinel-1 / EMIT acquisition over the boundary and how many were clear.	Planetary Computer, NASA CMR	Data availability
3	Crop type per year	Clips the national CDL to the boundary: dominant crop and its share, every year.	USDA CDL (local or CropScape)	CDL
4	Imagery and NDVI series	Reads only the boundary window of each Sentinel-2 scene; keeps scenes with 75%+ of field pixels clear; one field mean per date.	Sentinel-2 L2A (or HLS)	Imagery
5	Planting and harvest	Fits the NDVI curve each season; green-up minus a crop lag gives planting, the post-peak drop gives harvest; window widened by cloud gaps.	NDVI + CDL	Season analysis
6	Weather check	Moves each date off rain (and optionally wind) days and the day after heavy rain, within its window.	gridMET (CSIP)	Season analysis
7	Cover crops	Reads off-season NDVI between harvest and the next planting: verdict, seeding and termination dates.	NDVI + CDL	Cover crops
8	Soil armor	Splits each pixel into living, residue and bare; armor = 1 - bare, before planting and per calendar year.	Sentinel-2 SWIR	Soil armor
9	Radar + soil moisture	Reads Sentinel-1 radar steps (tillage, canopy removal) and a daily soil wetness estimate; narrows each planting and harvest window to dry-enough days.	Sentinel-1 RTC, gridMET	Soil moisture (radar)
10	CAI snapshots (Python only)	Cellulose Absorption Index on each clear EMIT scene: dry residue vs green canopy.	NASA EMIT (Earthdata)	Soil armor
11	Summary and export	One row per season with all of the above; CSVs and a report pack.	--	Summary

Also in the Python app: a **field study viewer** (results from the Field to Market project tests -- Illinois FLASH fields, county benchmark cohorts in Kansas, Illinois and Georgia, a New Mexico alfalfa field) and **HLS** imagery (Sentinel-2 + Landsat, 30 m) as an alternative to Sentinel-2.

1-3. Boundary, data availability and crop type

The boundary decides everything downstream: only pixels whose centers fall inside it are used. Before any analysis the tool shows what data exists over it and which crop CDL records each year.

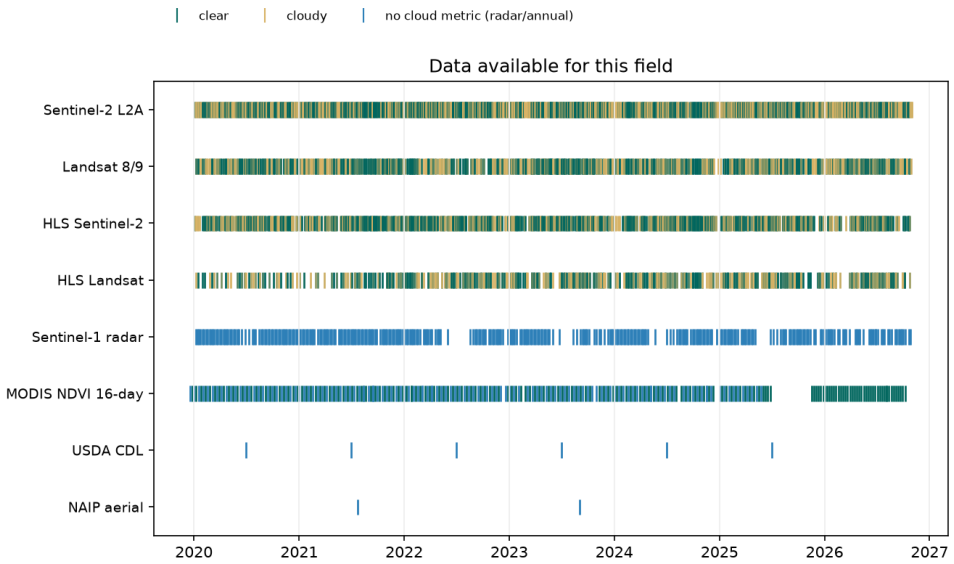


Figure 2. Data availability: every acquisition over the boundary by sensor and month.



Figure 3. CDL inside the boundary, one map per year: the crop rotation and how cleanly the boundary holds one crop.

4-6. NDVI series, planting and harvest, weather check

Each clear Sentinel-2 date gives one NDVI value for the field. Per season the curve is smoothed; planting is the green-up minus a crop-specific lag (e.g. corn), harvest is the drop after the peak. Shaded bands are the uncertainty windows, widened where clouds left gaps. With the weather check on, each date moves to the nearest workable day: dashed lines are the NDVI estimates, solid lines the weather-checked dates.

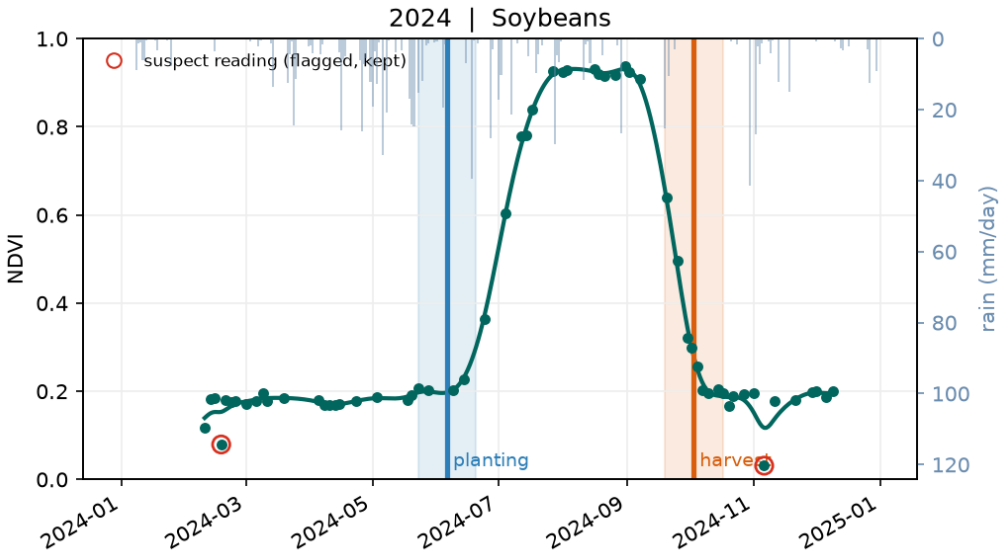


Figure 4. One season (2024, soybeans): NDVI points and curve, planting (blue) and harvest (orange) windows, rain bars from the top.

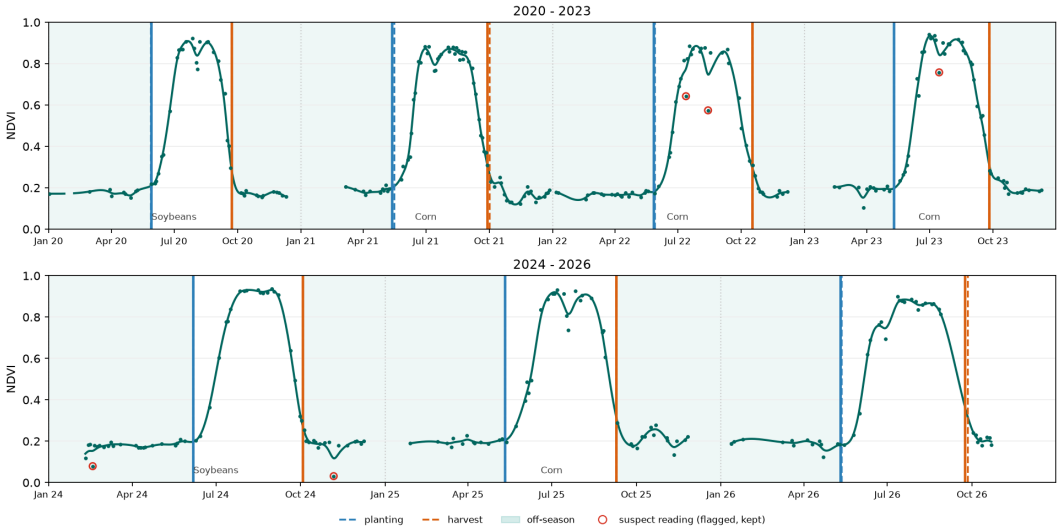


Figure 5. All seasons on one axis: the rotation, season lengths and off-season gaps.

7-8. Cover crops and soil armor

Cover crops are read from NDVI between harvest and the next planting: sustained green above 0.30 is a likely cover crop; a winter cash crop or perennial stand from CDL is named instead. **Soil armor** splits every pixel into living cover, crop residue and bare soil from NDVI and a shortwave dead-fuel index; armor = 1 - bare. It is reported for the pre-planting window and, since this release, as calendar-year residue and worst-case scores that do not favor winter crops.

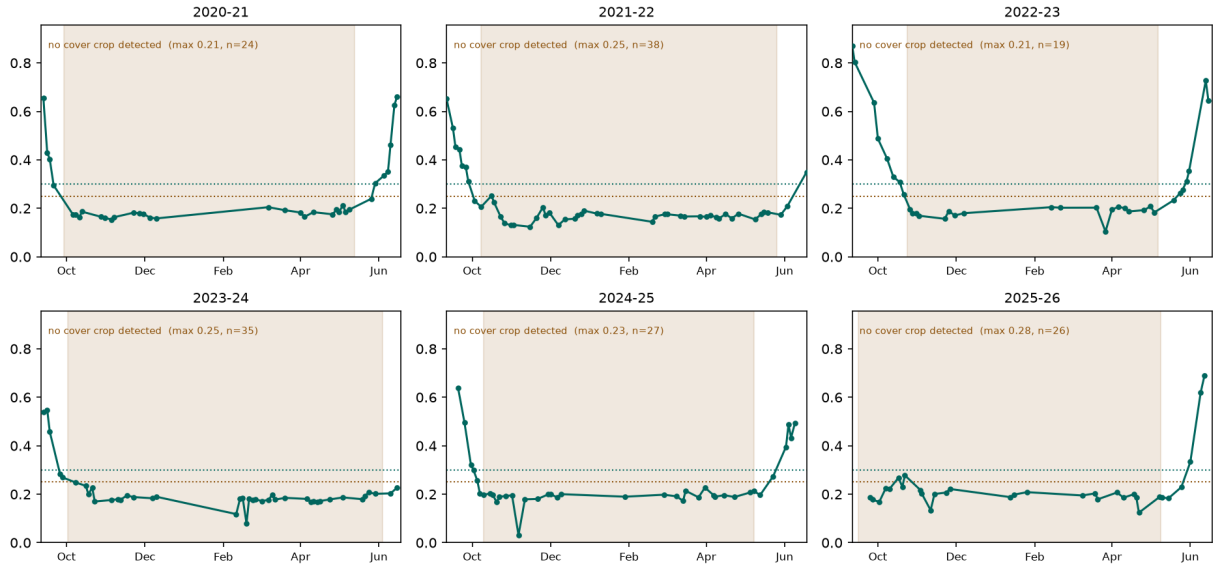


Figure 6. Off-season NDVI per winter with the cover crop verdict.

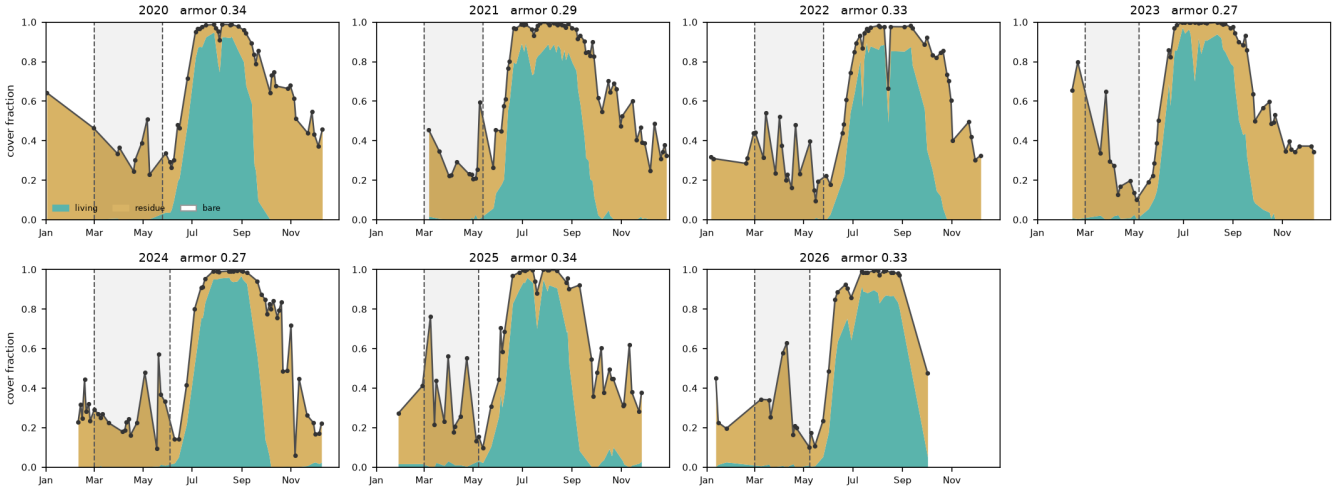


Figure 7. Soil armor per calendar year: living (green), residue (tan) and bare share, pre-planting window shaded.

9. Radar and soil moisture: narrowing the field-work dates

Sentinel-1 radar sees through cloud every ~6 days. A drop in VV backscatter on bare soil marks a seedbed or planter pass; a drop in VH marks the canopy being removed at harvest. Each NDVI window gets a radar interval: filled when it stands out from the field's noise, dotted when weak. Inside the window the tool keeps only days when the **soil is dry enough first** (daily wetness from rain memory, corrected by radar on bare passes) and **the weather second** (planting tolerates light rain; harvest needs a rain-free day). The tan strip marks those days; the bar under it is the narrowed window; the dotted line is the soil-moisture date shown beside the weather-checked date in the Summary.

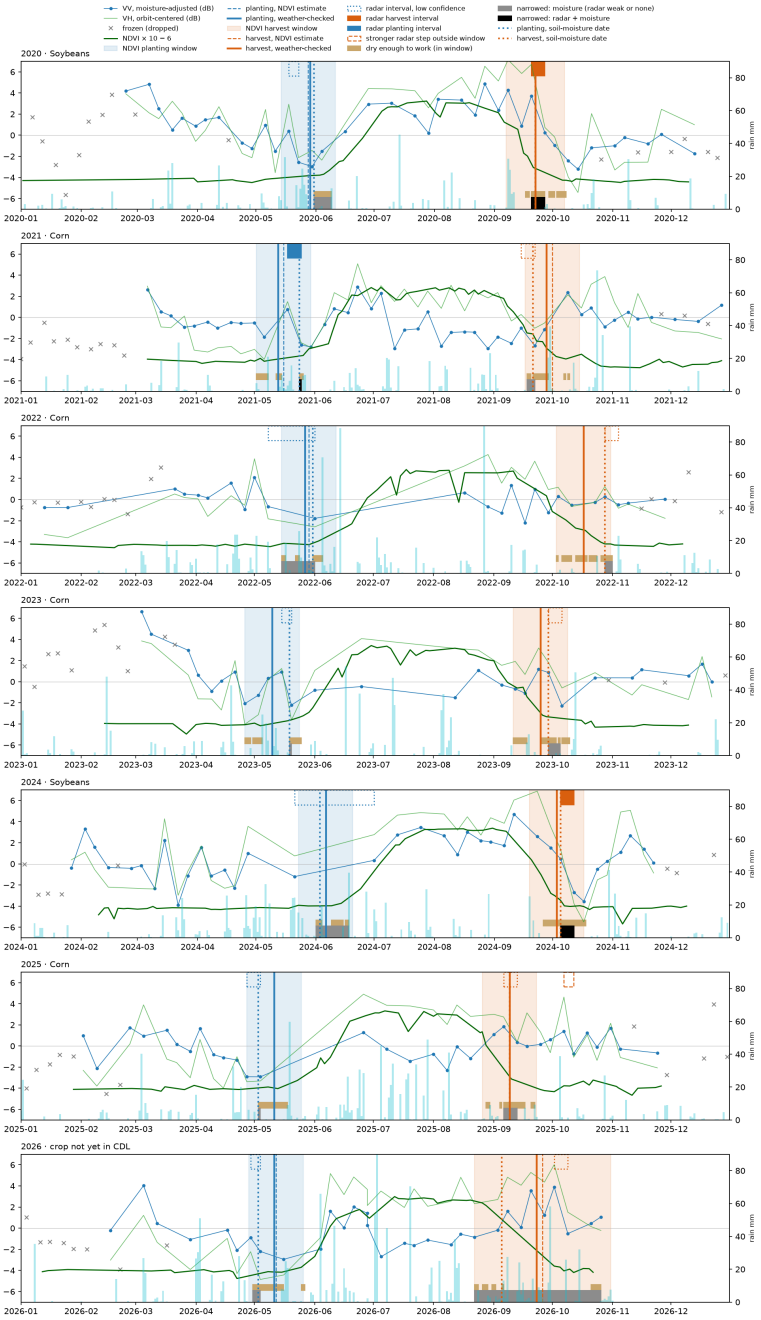


Figure 8. Radar against the NDVI windows, one panel per year.

10. Crop residue from hyperspectral snapshots (CAI, Python only)

The Cellulose Absorption Index reads the absorption of dry plant material near 2100 nm: CAI = 100 x (0.5 (R2031 + R2211) - R2101). Sentinel-2 cannot resolve it, so it comes from NASA EMIT on the Space Station (60 m, 2022 on). EMIT is a snapshot, not a time series: each clear scene gives one reading, and it means residue only where the ground is not green (NDVI below 0.30). Needs a free NASA Earthdata login.

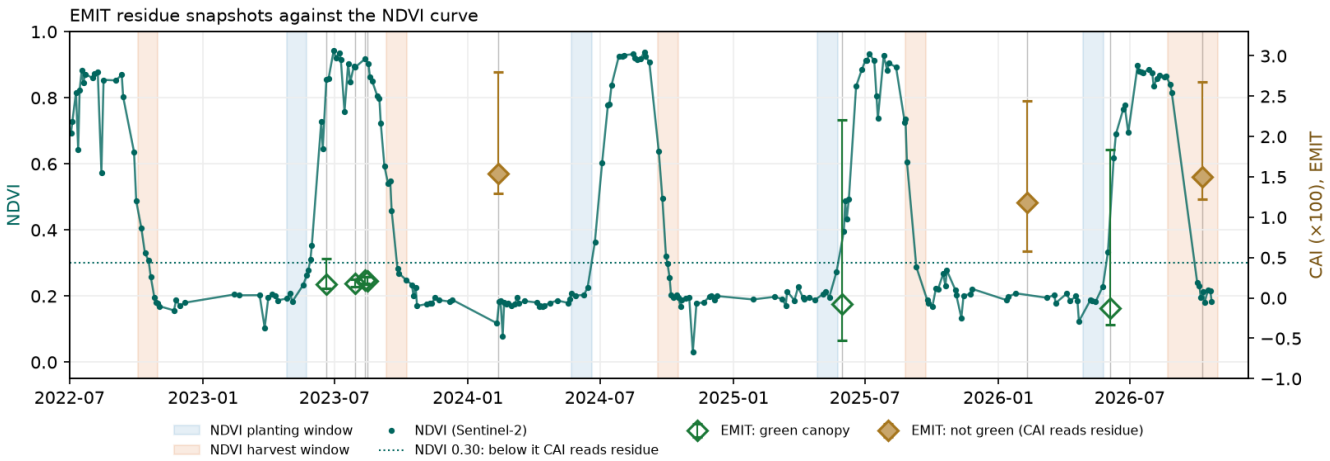


Figure 9. EMIT CAI (right axis) against the NDVI curve (left axis): tan = not green, CAI reads residue; hollow green = canopy.

9 usable of 14 scenes read for the example field:

Date	CAI (median)	NDVI	Reading
2023-06-20	0.17	0.88	green canopy
2023-07-30	0.18	0.93	green canopy
2023-08-12	0.22	0.92	green canopy
2023-08-16	0.21	0.91	green canopy
2024-02-12	1.54	0.14	dry residue
2025-05-31	-0.08	0.32	green canopy
2026-02-10	1.18	0.15	dry residue
2026-06-04	-0.13	0.45	green canopy
2026-10-09	1.50	0.24	dry residue

11. The Summary table

One row per season brings it together. Below, the example field: the weather-checked dates (shift from the NDVI estimate in days) beside the soil-moisture + radar dates, with the rule that set them.

Year	Crop	Planting (weather)	Planting (moisture)	Harvest (weather)	Harvest (moisture)	Following winter
2020	Soybeans	2020-05-29 (+1)	2020-05-31 (+3) moisture	2020-09-22 (+0)	2020-09-22 (+0) radar + moisture	no cover crop detected
2021	Corn	2021-05-13 (-3)	2021-05-24 (+8) radar + moisture	2021-09-28 (-3)	2021-09-21 (-10) radar (low) + moisture	no cover crop detected
2022	Corn	2022-05-27 (-2)	2022-05-31 (+2) radar (low) + moisture	2022-10-17 (+0)	2022-10-28 (+11) radar (low) + moisture	no cover crop detected
2023	Corn	2023-05-10 (+0)	2023-05-19 (+9) radar (low) + moisture	2023-09-25 (+0)	2023-09-29 (+4) radar (low) + moisture	no cover crop detected
2024	Soybeans	2024-06-06 (+0)	2024-06-03 (-3) radar (low) + moisture	2024-10-03 (+0)	2024-10-05 (+2) radar + moisture	no cover crop detected
2025	Corn	2025-05-11 (+0)	2025-05-03 (-8) radar (low) + moisture	2025-09-09 (+0)	2025-09-09 (+0) radar (low) + moisture	no cover crop detected
2026	--	2026-05-11 (-1)	2026-05-03 (-9) radar (low) + moisture	2026-09-23 (-3)	2026-09-05 (-21) moisture	--

Before quoting a number

Output	Checked against	Not yet checked
CDL crop	USDA accuracy reports	Boundary mixing two fields (flagged)
Planting / harvest	--	Grower-reported dates
Weather / moisture dates	--	Grower field-work records
Cover crop verdict	Georgia & Illinois fields: works in the Midwest, 7% specificity in Georgia	Regional thresholds
Soil armor	Consistent with cover crop verdicts	Line-transect cover
CAI (EMIT)	--	Residue cover measurements

Every threshold is a literature-typical starting value, documented with its source in METHODS.md. Read the outputs as a consistent screening record across fields, not as measurements of one field.